

Xuyuan Liu

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RESEARCH INTERESTS

My research focuses on **the science of Language Models**, with an emphasis on understanding their behavior during both training and inference by probing their internal mechanisms. My goal is to **understand how LLMs work internally** and to translate that understanding into technical methods that enhance their reasoning, reliability, and personalization capabilities.

EDUCATION

- **Dartmouth College** Sept. 2023 – Present
Ph.D. in Computer Science
◦ Advisor: Prof. [Yujun Yan](#)
Hanover, NH
- **Nankai University** Sept. 2019 – June 2023
B.Eng. in Computer Science
Tianjin, China

INTERNSHIP EXPERIENCE

- **Amazon** June 2026 – Sep. 2026
Applied Scientist Intern
Seattle, WA
- Developing an agentic search system that tackles the *credit assignment* problem in tool-use training, enabling more reliable reward attribution across multi-step tool invocations.
- **NEC Laboratories America** June 2025 – Sep. 2025
Research Scientist Intern
Princeton, NJ
Mentors: Dr. Zhengzhang Chen and Dr. Haifeng Chen
- Proposed a framework for *lifelong LLM memory customization* via hidden-state interventions, enabling precise, localized, and lightweight knowledge updates without full model retraining.
- **Outcome:** [CP.5] (Accepted to ACL 2026, [Oral](#)).

PUBLICATIONS & PREPRINTS

* EQUAL CONTRIBUTION; S=IN SUBMISSION, CP=CONFERENCE PROCEEDINGS

- [CP.5] Xuyuan Liu, Shengyu Chen, Xinshuai Dong, ..., Yujun Yan, Haifeng Chen, Zhengzhang Chen. **Representation Interventions Enable LLM's Lifelong Knowledge Memory Control**. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. [Oral](#).
💎 LLM Personalization 💎 LLM Memory Customization 💎 Efficient LLM
- [CP.4] Xuyuan Liu, Lei Hsiung, Yaoqing Yang, and Yujun Yan. **Spectral Insights into Data-Oblivious Critical Layers in Large Language Models**. In *Findings of the Association for Computational Linguistics (ACL 2025)*. Also presented at the *UniReps Workshop @ NeurIPS*.
💎 LLM Training Behavior 💎 Efficient LLM
- [CP.3] Xuyuan Liu, Yinghao Cai, Qihui Yang, and Yujun Yan. **Exploring Consistency in Graph Representations: From Graph Kernels to Graph Neural Networks**. In *Proceedings of the Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)*.
💎 Representation Learning
- [CP.2] Jinhang Chai, Xuyuan Liu, Elynn Chen, and Yujun Yan. **Low-Rank Plus Sparse Matrix Transfer Learning under Growing Representations and Ambient Dimensions**. In *Submission*.
💎 Representation Learning
- [CP.1] Zichen Liu, Xuyuan Liu, Yanlong Wen, Guoqing Zhao, Fen Xia, and Xiaojie Yuan. **TreeMAN: Tree-Enhanced Multimodal Attention Network for ICD Coding**. In *Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022)*. [Oral](#).
💎 Multi-Modality Learning

HONORS, AWARDS, AND GRANTS

- **Oral Presentation**. ACL 2026
- **Silver Reviewer Award**. ICML 2026
- **Guarini Travel Grant (2024)**. Guarini School of Graduate and Advanced Studies, Dartmouth College.
- **Dartmouth Fellowship (2023)**. Guarini School of Graduate and Advanced Studies, Dartmouth College.
- **Academic Excellence Scholarship**. Nankai University (Top 7/130).
- **Innovation Award and Research Scholarship**. Nankai University (Top 3/130).

TEACHING EXPERIENCE

Head Teaching Assistant, *Deep Learning* (Spring 2024); Head Teaching Assistant, *Machine Learning & Statistical Data Analysis* (Fall 2024), Dartmouth College.

PROFESSIONAL SERVICES

- **Reviewer:** ICLR 2025–2026, NeurIPS 2024–2025, ICML 2025–2026, KDD Dataset & Benchmark Track, TMLR, TPAMI

SKILLS

- **Programming Languages:** C, C++, Python, Java, SQL, Shell Scripting
- **ML/LLM Frameworks and Tools:** PyTorch, vLLM, HuggingFace, VeRL, OpenRLHF
- **Tools:** Git, $\LaTeX$, Slurm